

Appendix G - Per-CPU Opcode Quick Reference

One table per CPU. Each table is a single-line summary of every mnemonic; the full encoding, addressing modes, flag effects, and cycle counts are in the per-CPU chapter (Ch 24-29). This appendix is for the reader who has the chapter open already and only needs to look up "is this mnemonic spelled this way" or "what does this opcode do at a glance".

G.1 IE64 (Chapter 24)

Fixed 8-byte instruction, 32 GPRs R0-R31, R0 = 0, R31 = SP. Compare-and-branch ISA; no separate flag register.

Group	Mnemonics
ALU	ADD, SUB, MUL, DIV, MOD, AND, OR, XOR, NOT, NEG, SHL, SHR, SAR, ROL, ROR. Each has .Q (full register) and .L/.W/.B width variants for memory ops.
Load / store	LOAD.B/.W/.L/.Q, STORE.B/.W/.L/.Q. Address modes: register, register + immediate, register + register, PC-relative.
Immediate	MOVE.Q rd, #imm32, LI rd, #imm.
Branch	BRA, JSR, RTS, BEQ, BNE, BLT, BLE, BGT, BGE, BEQZ, BNEZ. All compare-and-branch forms take two GPRs and a target.
FPU	FADD, FSUB, FMUL, FDIV, FNEG, FABS, FSQRT, FCMP, FTOI, ITOF, single and double precision.
System	SYSCALL, BREAK, HALT, CRM0V (move to/from CR0-CR15), TLBI (invalidate one TLB entry), MFENCE.

G.2 IE32 (Chapter 25)

8-byte instruction, 16 named registers A,X,Y,Z,B,C,D,E,F,G,H,I,J,K,L,W, no MMU, no FPU.

Group	Mnemonics
ALU	ADD, SUB, MUL, DIV, AND, OR, XOR, NOT, NEG, SHL, SHR.
Memory	LOAD, STORE, with .B/.W/.L width suffixes.
Immediate	LDI A, #imm32.
Branch	JMP, JSR, RTS, JZ rA, JNZ rA (test single register against zero).
Stack	PUSH rA, POP rA.
Timer	direct MMIO (IE32_TIMER_COUNT, IE32_TIMER_PERIOD).

G.3 6502 (Chapter 26)

A, X, Y, S, P, PC. Addressing modes: implied, accumulator, immediate, zero-page (+ X, + Y), absolute (+ X, + Y), indirect, (indirect,X), (indirect),Y, relative.

Group	Mnemonics
Load / store	LDA, LDX, LDY, STA, STX, STY.
Transfer	TAX, TAY, TXA, TYA, TSX, TXS.
Arith / log	ADC, SBC, AND, ORA, EOR, BIT, CMP, CPX, CPY.

Group	Mnemonics
Inc / dec	INC, DEC, INX, INY, DEX, DEY.
Shift	ASL, LSR, ROL, ROR.
Branch	BCC, BCS, BEQ, BNE, BMI, BPL, BVC, BVS.
Jump / sub	JMP, JSR, RTS, RTI.
Flag	CLC, SEC, CLD, SED, CLI, SEI, CLV.
Stack	PHA, PLA, PHP, PLP.
System	BRK, NOP.
Undoc	LAX, SAX, DCP, ISC, RLA, RRA, SLO, SRE, ANC, ARR, ASR, AXS, XAA.

Flag register P: bit 0 C, 1 Z, 2 I, 3 D, 4 B, 5 -, 6 V, 7 N (silicon convention; see AUTHOR_PROVENANCE for the MOS 1976 manual divergence).

G.4 Z80 (Chapter 27)

Main: A, F, B, C, D, E, H, L; alt: A', F', B', C', D', E', H', L'; index: IX, IY; refresh / interrupt: I, R; stack: SP. Prefixes: CB (bit / rotate), ED (extended), DD / FD (IX / IY).

Group	Mnemonics
8-bit load	LD r, r', LD r, n, LD r, (HL), LD r, (IX+d), LD A, (BC)/(DE)/(nn).
16-bit load	LD rp, nn, LD rp, (nn), LD (nn), rp, PUSH rp, POP rp.
Block	LDI, LDIR, LDD, LDDR, CPI, CPIR, CPD, CPDR.
Arith / log	ADD, ADC, SUB, SBC, AND, OR, XOR, CP, INC, DEC.
16-bit ALU	ADD HL, rp, ADC HL, rp, SBC HL, rp, ADD IX, rp, ADD IY, rp.
Rotate / shift	RLCA, RLA, RRCA, RRA, RLC, RL, RRC, RR, SLA, SRA, SRL, SLL (undoc).
Bit	BIT b, r, SET b, r, RES b, r.
Jump / call	JP, JR, CALL, RET, with condition codes NZ, Z, NC, C, PO, PE, P, M. DJNZ.
I/O	IN A, (n), IN r, (C), OUT (n), A, OUT (C), r, INI, OUTI, INIR, OTIR, IND, OUTD, INDR, OTDR.
Interrupt	EI, DI, IM 0, IM 1, IM 2, RETI, RETN.
Misc	NOP, HALT, EX, EXX, DAA, CPL, NEG, CCF, SCF.

Flag register: S, Z, Y (bit 5 undoc), H, X (bit 3 undoc), P/V, N, C.

G.5 M68K 68020 (Chapter 28)

D0-D7, A0-A7, PC, SR. CCR low byte: X N Z V C. Big-endian, byte-addressable, 32-bit address bus.

Group	Mnemonics
Move	MOVE, MOVEA, MOVEM, MOVEP, MOVEQ, EXG, SWAP, LEA, PEA.

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Group	Mnemonics
Arith	ADD, ADDA, ADDI, ADDQ, ADDX, SUB, SUBA, SUBI, SUBQ, SUBX, NEG, NEGX, CMP, CMPA, CMPI, CMPM, CLR, EXT, EXTB.
Mul / div	MULU, MULS, DIVU, DIVS (68020 also MULU.L, MULS.L, DIVUL, DIVSL).
Logical	AND, ANDI, OR, ORI, EOR, EORI, NOT.
Shift / rot	ASL, ASR, LSL, LSR, ROL, ROR, R0XL, R0XR.
Bit	BTST, BSET, BCLR, BCHG.
Bit field	BFTST, BFSET, BFCLR, BFCHG, BFEXTS, BFEXTU, BFINS, BFFF0.
Branch	BRA, BSR, Bcc (HI, LS, CC, CS, NE, EQ, VC, VS, PL, MI, GE, LT, GT, LE), DBcc, Scc, JMP, JSR, RTS, RTR, RTD.
BCD	ABCD, SBCD, NBCD.
Multiprocessor	TAS, CAS, CAS2.
System	TRAP #n, TRAPV, CHK, CHK2, STOP, RESET, ILLEGAL, NOP, LINK, UNLK, MOVE USP, MOVEC, MOVES, BKPT, RTE.
Line A/F	unassigned opcode trap.

G.6 x86 (Chapter 29, 8086 base + 386 extensions)

EAX, EBX, ECX, EDX, ESI, EDI, EBP, ESP, plus 16-bit and 8-bit subregisters. Segments CS, DS, ES, FS, GS, SS. Real-mode only.

Group	Mnemonics
Move	MOV, MOVZX, MOVZX, LEA, XCHG, XLAT, PUSH, POP, PUSHA, POPA, PUSHAD, POPAD, PUSHF, POPF, PUSHFD, POPFD.
Arith	ADD, ADC, SUB, SBB, INC, DEC, CMP, NEG, MUL, IMUL, DIV, IDIV, DAA, DAS, AAA, AAS, AAM, AAD, CBW, CWD, CWDE, CDQ.
Logical	AND, OR, XOR, NOT, TEST.
Shift / rot	SHL, SHR, SAR, SAL, ROL, ROR, RCL, RCR, SHLD, SHRD.
Bit	BT, BTS, BTR, BTC, BSF, BSR, SETcc.
Branch	JMP, Jcc (JE, JNE, JL, JG, JLE, JGE, JB, JA, JBE, JAE, JO, JNO, JS, JNS, JP, JNP, JCXZ, JECXZ), CALL, RET, RETF, LOOP, LOOPE, LOOPNE, INT n, INTO, IRET.
String	MOVS, CMPS, SCAS, LODS, STOS, INS, OUTS, prefixes REP, REPE, REPNE.
I/O	IN, OUT.
Flag	STC, CLC, CMC, STD, CLD, STI, CLI, LAHF, SAHF.
Segment	LDS, LES, LFS, LGS, LSS.
System	HLT, WAIT, NOP, ESC, LOCK.
386 extras	BSWAP, MOVSD, dword forms of all 16-bit ops via 66h / 67h prefixes.

Omitted (Chapter 29): all protected-mode opcodes (LGDT, LIDT, LLDT, LTR, LMSW, SMSW, ARPL, LAR, LSL, VERR, VERW, STR, SLDT), CR and DR register moves, INVLPG, WBINVD, INVD.